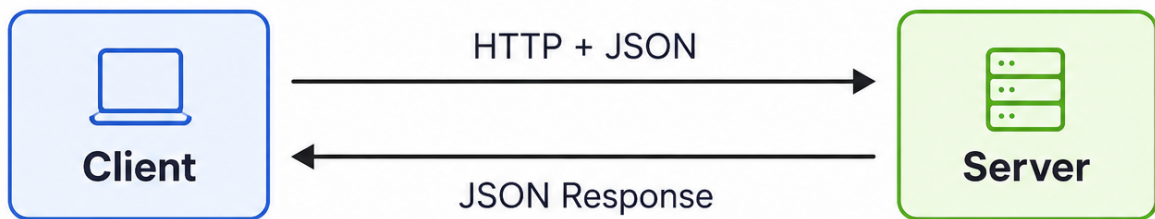


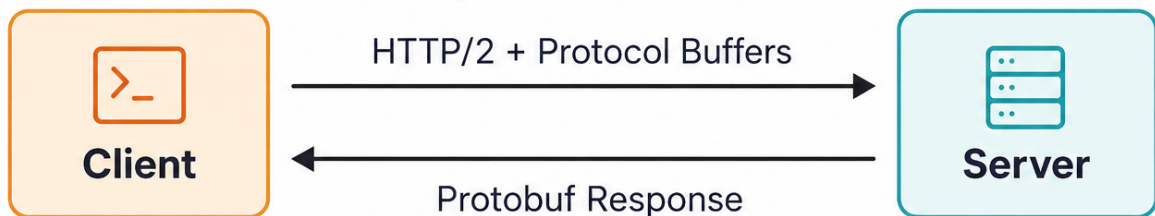
Backend diagrams

REST + gRPC APIs

REST API



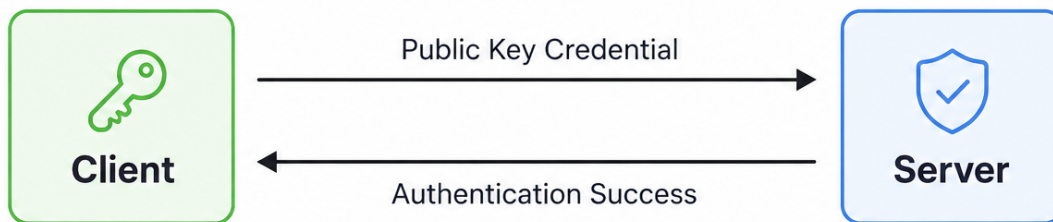
gRPC API



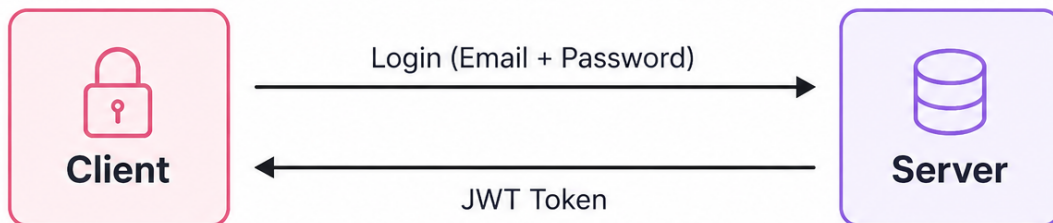
MODERN AUTH

PASSKEYS > JWT

PASSKEYS

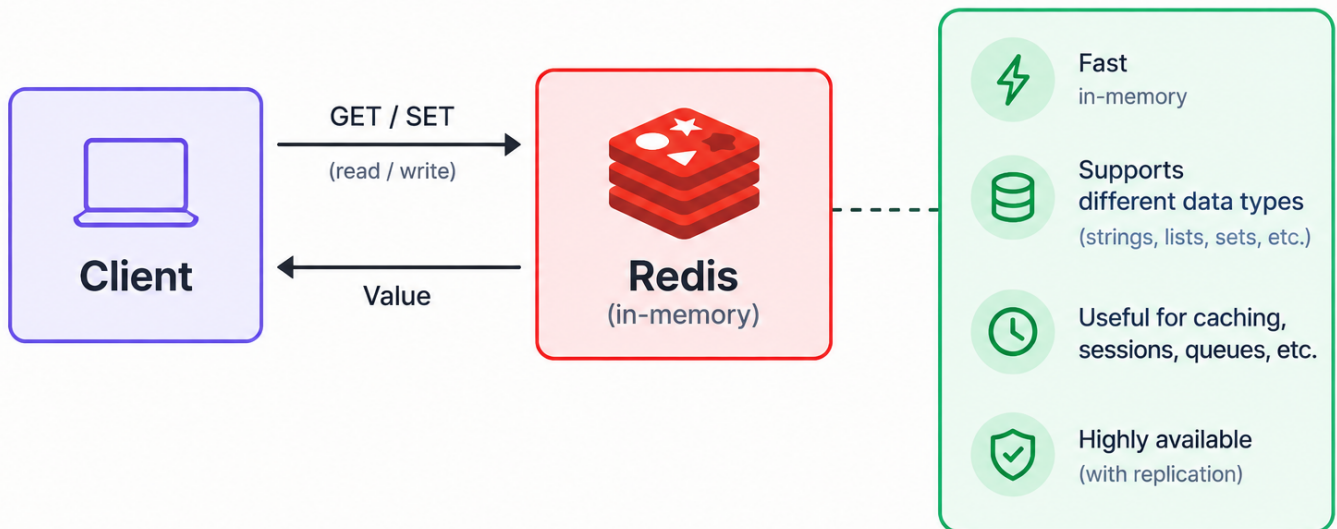


JWT (LEGACY)



REDIS

A fast, in-memory data store for caching, queues and more.



SQL vs NoSQL

SQL



Relational Database
Structured data

ID	Name	Email	Age
1	Ana	ana@mail.com	25
2	Luis	luis@mail.com	30
3	Mia	mia@mail.com	28

- ✓ Fixed schema (tables)
- ✓ ACID transactions
- ✓ Powerful joins
- ✓ Best for complex queries

VS

NoSQL



Non-Relational Database
Flexible data

```
{
  "id": 1,
  "name": "Ana",
  "email": "ana@mail.com",
  "age": 25,
  "skills": ["JS", "React"]
}
```

- ✓ Flexible schema (documents)
- ✓ Horizontal scaling
- ✓ High performance
- ✓ Best for large scale / evolving data

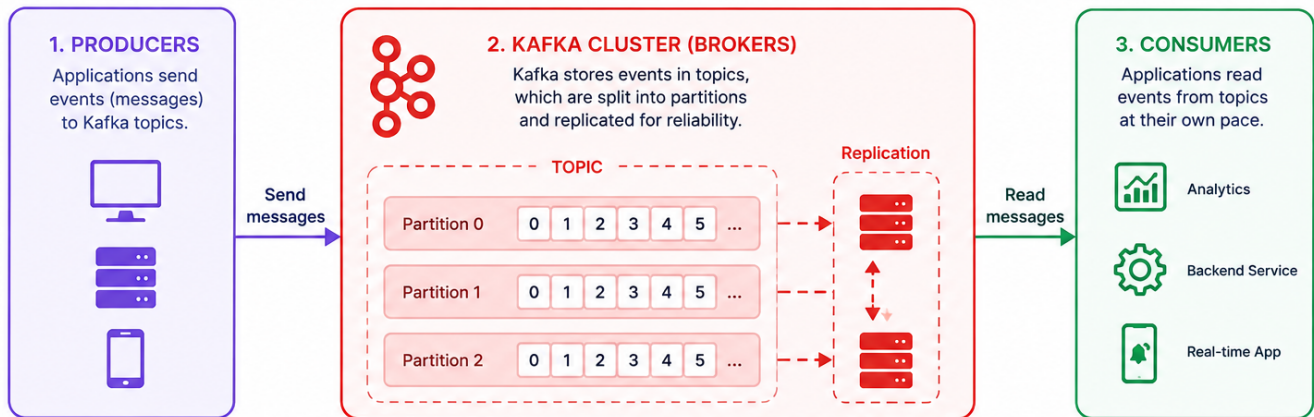


Choose **SQL** when you need structure and complex relationships.

Choose **NoSQL** when you need flexibility and scale.

KAFKA

A distributed event streaming platform for building real-time data pipelines and applications.



KEY BENEFITS



High Throughput
Handles millions of messages per second.



Durable & Reliable
Messages are persisted and replicated.



Scalable
Scale horizontally as your data grows.



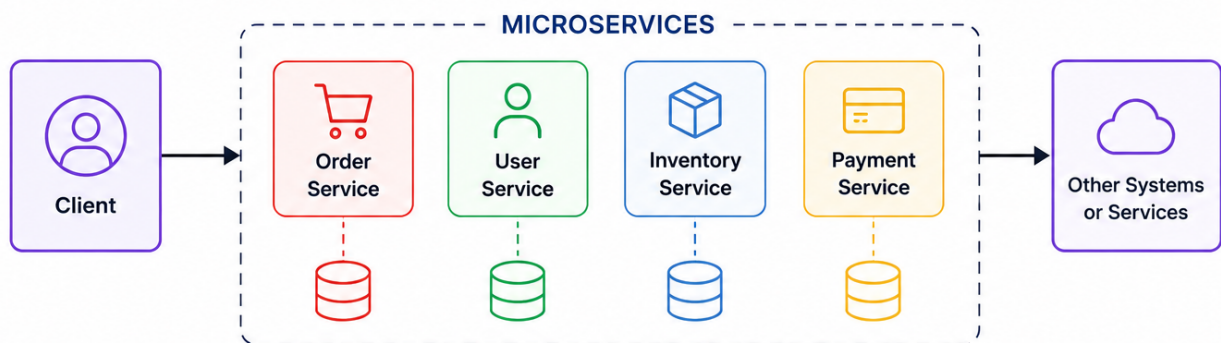
Real-time
Process and analyze data in real time.



IN SHORT: Producers send data to Kafka topics. Kafka stores it reliably across partitions and brokers. Consumers read the data when they need it.

MICROSERVICES

An architectural style that structures an application as a collection of small, independent services, each focused on a specific business capability.



KEY BENEFITS

-  **Independent**
Each service is developed, deployed and scaled independently.
-  **Scalable**
Scale only the services that need more capacity.
-  **Resilient**
Failure in one service does not affect the others.
-  **Agile**
Teams can deliver changes faster and iterate independently.



IN SHORT: Microservices break down an application into small services that communicate through APIs, making the system more flexible, scalable and easier to maintain.